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Semiconductor package

Abstract

A semiconductor package includes a substrate, a redistribution circuit layer, and a protective layer. The redistribution circuit layer is over the substrate and includes a plurality of functional pads electrically connected to the substrate, and a dummy pad pattern electrically disconnected from the plurality of functional pads, wherein the dummy pad pattern includes a plurality of pad portions connected to one another. The protective layer is disposed over the redistribution circuit layer and comprising a plurality of first openings spaced apart from one another and respectively revealing the plurality of pad portions.

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Background/Summary

BACKGROUND

(1) The semiconductor industry has experienced rapid growth due to ongoing improvements in the integration density of a variety of electronic components (e.g., transistors, diodes, resistors, capacitors, etc.). For the most part, improvement in integration density has resulted from iterative reduction of minimum feature size, which allows more components to be integrated into a given area. As the demand for shrinking electronic devices has grown, a need for smaller and more creative packaging techniques of semiconductor dies has emerged. An example of such packaging systems is Package-on-Package (PoP) technology. In a PoP device, a top semiconductor package is stacked on top of a bottom semiconductor package to provide a high level of integration and component density. PoP technology generally enables production of semiconductor devices with enhanced functionalities and small footprints on a printed circuit board (PCB).

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

(2) FIG. 1 to FIG. 11 illustrate partial cross sectional views of intermediate stages in the manufacturing of a semiconductor package according to some embodiments of the present disclosure.

(3) FIG. 12 illustrates a partial top view of an intermediate stage in the manufacturing of a semiconductor package according to some embodiments of the present disclosure.

(4) FIG. 13 illustrates a partial enlarged view of the semiconductor package in FIG. 12.

(5) FIG. 14 illustrates a schematic cross sectional view of the semiconductor package in FIG. 13.

(6) FIG. 15 illustrates a partial enlarged view of the semiconductor package in FIG. 12.

(7) FIG. 16 illustrates a schematic cross sectional view of the semiconductor package in FIG. 15.

(8) FIG. 17 illustrates a partial enlarged view of the semiconductor package in FIG. 12.

DETAILED DESCRIPTION

(9) The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

(10) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

(11) Embodiments discussed herein may be discussed in a specific context, namely a

semiconductor package (e.g., a package on package (PoP) structure) having dummy connectors disposed between a first package and a second package component. In some embodiment, the second package component may be a memory package (e.g., having one or more dynamic random access memory (DRAM) dies), which is physically and electrically coupled to the first package component (e.g., an integrated fan-out (InFO) package having a logic die and redistribution structures) using functional connectors. The dummy connectors may be physically and/or electrically disconnected from functional circuits of the first package component (e.g., the logic die and the redistribution structures). In some embodiments, the dummy connectors may also be electrically disconnected from functional circuits of the second package component.

(12) Accordingly, the redistribution structure of the first package component may include corresponding dummy pads for connecting the dummy connectors. The dummy pads may also be used to provide a uniform metal distribution within the redistribution structure and homogenize the density of metal features in the package component, preventing irregularities in the surface. Thus, subsequent structures are formed on a uniform and planar surface. However, dummy pads are isolated from other functional conductive features such as functional pads, functional circuits, etc., so heat is harder to dissipate during thermal process such as laser drilling process, or the like. Hence, in some embodiments, a plurality of dummy pads can be connected to one another to be integrated into single dummy pad pattern for improving heat dissipation efficiency.

(13) It is noted that the teachings of this disclosure are applicable to any package structure including functional connectors bonding two package components. Other embodiments contemplate other applications, such as different package types or different configurations that would be readily apparent to a person of ordinary skill in the art upon reading this disclosure. It should be noted that embodiments discussed herein may not necessarily illustrate every component or feature that may be present in a structure. For example, multiples of a component may be omitted from a figure, such as when discussion of one of the component may be sufficient to convey aspects of the embodiment. Further, method embodiments discussed herein may be discussed as being performed in a particular order; however, other method embodiments may be performed in any logical order.

(14) FIG. 1 to FIG. 11 illustrate partial cross sectional views of intermediate stages in the manufacturing of a semiconductor package according to some embodiments of the present disclosure. FIG. 1 illustrates a carrier substrate **100** and a release layer **102** formed over the carrier substrate **100**. In some embodiments, a plurality of package regions **600** and **602** for the formation of a plurality of packages respectively, are illustrated. The number of the package regions are merely for illustration and is not limited in the disclosure.

(15) The carrier substrate **100** may be a glass carrier substrate, a ceramic carrier substrate, or the like. The carrier substrate **100** may be a wafer, such that multiple packages can be formed on the carrier substrate **100** simultaneously. The release layer **102** may be formed of a polymer-based material, which may be removed along with the carrier substrate **100** from the overlying structures that will be formed in subsequent steps. In some embodiments, the release layer **102** is an epoxy-based thermal-release material, which loses its adhesive property when heated, such as a light-to-heat-conversion (LTHC) release coating. In other embodiments, the release layer **102** may be a ultra-violet (UV) glue, which loses its adhesive property when exposed to UV lights. The release layer **102** may be dispensed as a liquid and cured, may be a laminate film laminated onto the carrier substrate **100**, or may be the like. The top surface of the release layer **102** may be leveled and may have a high degree of coplanarity.

(16) Referring to FIG. 2, a protective layer **104** and a redistribution circuit layer including functional pads **106** and dummy pad pattern **107** are formed. As illustrated in FIG. 2, a protective layer **104** is formed on the release layer **102**. The bottom surface of the protective layer **104** may be in contact with the top surface of the release layer **102**. In some embodiments, the protective layer **104** is formed of a polymer, which may be a photo-sensitive material such as polybenzoxazole

(PBO), polyimide, benzocyclobutene (BCB), or the like, that may be patterned using a lithography mask. In other embodiments, the protective layer **104** is formed of a nitride such as silicon nitride; an oxide such as silicon oxide, phosphosilicate glass (PSG), borosilicate glass (BSG), boron-doped phosphosilicate glass (BPSG), or the like; or the like. The protective layer **104** may be formed by any acceptable deposition process, such as spin coating, chemical vapor deposition (CVD), laminating, sputtering, the like, or a combination thereof.

(17) In some embodiments, the redistribution circuit layer **106, 107** is formed over the protective layer **104**. As an example to form redistribution circuit layer, a seed layer (not shown) is formed over the protective layer **104**. In some embodiments, the seed layer is a metal layer, which may be a single layer or a composite layer comprising a plurality of sub-layers formed of different materials. In some embodiments, the seed layer comprises a titanium layer and a copper layer over the titanium layer. The seed layer may be formed using, for example, PVD or the like. A photo resist is then formed and patterned on the seed layer. The photo resist may be formed by spin coating or the like and may be exposed to light for patterning. The pattern of the photo resist corresponds to the metallization pattern **106**. The patterning forms openings through the photo resist to expose the seed layer. A conductive material is formed in the openings of the photo resist and on the exposed portions of the seed layer. The conductive material may be formed by plating, such as electroplating or electroless plating, or the like. The conductive material may comprise a metal, like copper, titanium, tungsten, aluminum, or the like. Then, the photo resist and portions of the seed layer on which the conductive material is not formed are removed. The photo resist may be removed by an acceptable ashing or stripping process, such as using an oxygen plasma or the like. Once the photo resist is removed, exposed portions of the seed layer are removed, such as by using an acceptable etching process, such as by wet or dry etching. The remaining portions of the seed layer and conductive material form the redistribution circuit layer **106, 107**.

(18) In some embodiments, referring to as illustrated in FIG. 2, the redistribution circuit layer includes a plurality of functional pads **106** and at least one dummy pad pattern **107** electrically disconnected from the functional pads **106**. In some embodiments, the dummy pad pattern **107** is floating. The dummy pad pattern **107** are configured to provide a mounting point for subsequent formation of dummy connectors (e.g., dummy connectors **316** shown in FIG. 11. One of the examples of a top view of the redistribution circuit layer is illustrated in FIG. 12. In the present embodiment, a plurality of circular dummy pads are merged into one dummy pad pattern **107** as shown in regions A1, A3 in FIG. 12. In other words, a plurality of circular dummy pad portions are connected to one another by connecting portions to form the dummy pad pattern **107**. Accordingly, a size (e.g., footprint) of the dummy pad pattern **107** may be substantially greater than a size (e.g., footprint) of one of the functional pads **106**. More details on the dummy pad pattern **107** are described later in the disclosure.

(19) Referring to FIG. 3, a dielectric layer **108** is formed on the redistribution circuit layer **106, 107** and the protective layer **104**. In some embodiments, the dielectric layer **108** is formed of a polymer, which may be a photo-sensitive material such as PBO, polyimide, BCB, or the like, that may be patterned using a lithography mask. In other embodiments, the dielectric layer **108** is formed of a nitride such as silicon nitride; an oxide such as silicon oxide, PSG, BSG, BPSG; or the like. The dielectric layer **108** may be formed by spin coating, lamination, CVD, the like, or a combination thereof. The dielectric layer **108** is then patterned to form openings **1081** to reveals the functional pads **106** respectively for electrical connection. The patterning may be by an acceptable process, such as by exposing the dielectric layer **108** to light when the dielectric layer is a photo-sensitive material or by etching using, for example, an anisotropic etch. In the current embodiments, the dielectric layer **108** covers the dummy pad pattern **107**. In an alternative embodiment, the dielectric layer **108** may further include openings for exposing a part of the dummy pad pattern **107**.

(20) The dielectric layers **104** and **108** and the redistribution circuit layer **106, 107** may be referred to as a backside redistribution structure **110**. As illustrated, the backside redistribution structure **110**

includes the two dielectric layers **104** and **108** and one redistribution circuit layer **106**, **107**. In other embodiments, the back-side redistribution structure **110** can include any number of dielectric layers, redistribution circuit layers, and vias. One or more additional redistribution circuit layer and dielectric layer may be formed in the backside redistribution structure **110** by repeating the processes for forming a redistribution circuit layer and dielectric layer. Vias may be formed during the formation of a metallization pattern by forming the seed layer and conductive material of the metallization pattern in the opening of the underlying dielectric layer. The vias may therefore interconnect and electrically couple the various metallization patterns. The redistribution circuit layer including the functional pads **106** and the dummy pads **107** are formed as the first (outermost) metallization layer of the backside redistribution structure **110**.

(21) With now reference to FIG. 4, a plurality of through vias **112** are formed. As an example to form the through vias **112**, a seed layer is formed over the back-side redistribution structure **110**, e.g., the dielectric layer **108** and the functional pads **106** exposed by the openings **1081** of the dielectric layer **108** as illustrated. In some embodiments, the seed layer is a metal layer, which may be a single layer or a composite layer comprising a plurality of sub-layers formed of different materials. In some embodiments, the seed layer comprises a titanium layer and a copper layer over the titanium layer. The seed layer may be formed using, for example, PVD or the like. A photo resist is formed and patterned on the seed layer. The photo resist may be formed by spin coating or the like and may be exposed to light for patterning. The pattern of the photo resist corresponds to through vias. The patterning forms openings through the photo resist to expose the seed layer. A conductive material is formed in the openings of the photo resist and on the exposed portions of the seed layer. The conductive material may be formed by plating, such as electroplating or electroless plating, or the like. The conductive material may comprise a metal, like copper, titanium, tungsten, aluminum, or the like. The photo resist and portions of the seed layer on which the conductive material is not formed are removed. The photo resist may be removed by an acceptable ashing or stripping process, such as using an oxygen plasma or the like. Once the photo resist is removed, exposed portions of the seed layer are removed, such as by using an acceptable etching process, such as by wet or dry etching. The remaining portions of the seed layer and conductive material form through vias **112**. In some embodiments, the through vias **112** are functional through vias which are electrically connected to the functional pads **106**. In the embodiment of the dielectric layer **108** exposing a part of the dummy pad pattern **107**, at least one dummy through via (e.g., dummy through via **113** shown in FIG. 12 and FIG. 16) may be formed on the part of the dummy pad pattern **107** that is exposed by the dielectric layer **108**. As illustrated in FIG. 12, the position of the dummy through via **113** at least partially overlap with the dummy pad pattern **107** from a top view.

(22) Referring to FIG.-4 5, a plurality of integrated circuit dies **114** are adhered to the dielectric layer **108** by an adhesive **116**. As illustrated in FIG.-4 5 one integrated circuit die **114** is adhered in each of the package regions **600** and **602**, and in other embodiments, more integrated circuit dies **114** may be adhered in each region. For example, in an embodiment, two integrated circuit dies **114** may be adhered in each region. The integrated circuit die **114** may be a logic die (e.g., central processing unit, microcontroller, etc.), or the like. In some embodiments, the die **114** is a system on chip (SOC) that integrates all electronic components into a single die. In some embodiments, the die **114** may a die, a chip or a package. In some embodiments, the die **114** includes any one of various known types of semiconductor devices such as a memory die (e.g., dynamic random access memory (DRAM) die, static random access memory (SRAM) die, etc.), application-specific integrated circuit (ASIC) die, power management die (e.g., power management integrated circuit (PMIC) die), radio frequency (RF) die, sensor die, micro-electro-mechanical-system (MEMS) die, signal processing die (e.g., digital signal processing (DSP) die), front-end die (e.g., analog front-end (AFE) die), the like, or a combination thereof. In some embodiments, the die **114** has a top cross section (a cross section from the top view of the semiconductor package) in a quadrilateral, a

rectangular or a square shape.

(23) Before being adhered to the dielectric layer **108**, the die **114** may be processed according to applicable manufacturing processes to form integrated circuits in the die **114**. For example, the die **114** includes a semiconductor substrate **118**, such as silicon, doped or undoped, or an active layer of a semiconductor-on-insulator (SOI) substrate. The semiconductor substrate may include other semiconductor material, such as germanium; a compound semiconductor including silicon carbide, gallium arsenic, gallium phosphide, indium phosphide, indium arsenide, and/or indium antimonide; an alloy semiconductor including SiGe, GaAsP, AlInAs, AlGaAs, GaInAs, GaInP, and/or GaInAsP; or combinations thereof. Other substrates, such as multi-layered or gradient substrates, may also be used. Devices, such as transistors, diodes, capacitors, resistors, etc., may be formed in and/or on the semiconductor substrate **118** and may be interconnected by interconnect structures **120** formed by, for example, metallization patterns in one or more dielectric layers on the semiconductor substrate **118** to form an integrated circuit.

(24) In some embodiments, the die **114** further includes pads **122**, such as aluminum pads, to which external connections are made. The pads **122** are on what may be referred to as respective active surface (e.g., front side) of the die **114**. Passivation films **124** are on the die **114** and on portions of the pads **122**. Openings are through the passivation films **124** to the pads **122**. Die connectors **126**, such as conductive pillars (for example, comprising a metal such as copper), are in the openings through passivation films **124** and are mechanically and electrically coupled to the respective pads **122**. The die connectors **126** may be formed by, for example, plating, or the like. The die connectors **126** electrically couple the respective integrated circuits of the die **114**. The die connectors **126** are disposed on the active surface (e.g., front side) of the die **114**, and the backside redistribution structure **110** faces the back surface (e.g., backside) of the die **114**, which is opposite to the active surface of the die **114**.

(25) A dielectric material **128** is on the active surface of the die **114**, such as on the passivation films **124** and the die connectors **126**. The dielectric material **128** laterally encapsulates the die connectors **126**, and the dielectric material **128** is laterally coterminous with the respective integrated circuit dies **114**. The dielectric material **128** may be a polymer such as PBO, polyimide, BCB, or the like; a nitride such as silicon nitride or the like; an oxide such as silicon oxide, PSG, BSG, BPSG, or the like; the like, or a combination thereof, and may be formed, for example, by spin coating, lamination, CVD, or the like.

(26) In some embodiments, the adhesive **116** is attached to a backside of the die **114** and adheres the die **114** to the backside redistribution structure **110**, such as the dielectric layer **108** in the illustration. Accordingly, the adhesive **116** is located between the die **114** and the backside redistribution structure **110**. The adhesive **116** may be any suitable adhesive, epoxy, die attach film (DAF), or the like. The adhesive **116** may be applied to a backside of the dies **114**, such as to a backside of the respective semiconductor wafer or may be applied over the surface of the carrier substrate **100**. The dies **114** may be singulated, such as by sawing or dicing, and adhered to the dielectric layer **108** by the adhesive **116** using, for example, a pick-and-place tool.

(27) Referring now to FIG. 4 6 an encapsulating material **130** is formed over the dielectric layer **108** to at least laterally encapsulate the die **114** and the through vias **112** (and dummy through vias **113**, if applicable). In some embodiments, the molding process is a wafer-level molding process. For example, the encapsulating material **130** is dispensed to fill gaps between the die **114** and the through vias **112**. The encapsulating material **130** may include any suitable material such as molding compound, epoxy resin, molding underfill, or the like. Suitable methods for forming the encapsulating material **130** may include compressive molding, transfer molding, liquid encapsulant molding, and the like. For example, the encapsulating material **130** may be dispensed between the die **114** and the through vias **112** in liquid form. Subsequently, a curing process is performed to solidify the encapsulating material **130**. The filling of the encapsulating material **130** may overflow the die **114** and the through vias **112** so that the encapsulating material **130** covers top surfaces of

the die **114** and the through vias **112**. Accordingly, after curing, a mechanical grinding, chemical mechanical polish (CMP), or other etch back technique may be employed to remove excess portions of the encapsulating material **130** and expose the connectors **126** of the die **114** and the through vias **112** (dummy through vias **113**, if applicable). After planarization, top surfaces of the encapsulating material **130**, the die **114**, and the through vias **112** may be substantially level. In some embodiments, the grinding may be omitted, for example, if through vias **112** (dummy through vias **113**, if applicable) and die connectors **126** are already exposed. Accordingly, a plurality of through vias **112** (and dummy through vias **113**, if applicable) extending through the encapsulating material **130**. The through vias **112** are electrically connected to the die **114**.

(28) Throughout the description, the resultant structure including the die **114**, the through vias **112** and **113** (if any) and the encapsulating material **130** as shown in FIG. **6** is referred to as an encapsulated semiconductor device **101**, which may have a wafer form in the process. In some embodiments, the encapsulated semiconductor device **101** may be seen as a substrate **101**. Accordingly, in the encapsulated semiconductor device **101**, the encapsulating material **130** laterally encapsulates the die **114** and the through vias **112** and **113** (if any), the through vias **112** and **113** (if any) extend through the encapsulated semiconductor device **101** outside the die area where the die **114** is disposed. In other words, the encapsulating material **130** laterally encapsulates the die **110** therein, and the through vias **112** and **113** (if any) extends through the encapsulating material **130**. From a structural point of view, as illustrated in FIG. **6**, the backside redistribution structure **110** is disposed over a backside of the die **114** and the encapsulating material **130**. That is, the backside redistribution structure **110** is disposed over a backside of the encapsulated semiconductor device **101**.

(29) Referring to FIG. **7**, a front side redistribution structure **160** is formed. The front side redistribution structure **160** is disposed over a front side of the encapsulated semiconductor device **101**, e.g., the front side of the die **114**, the through vias **112/113** and the encapsulating material **130**, etc. The front side redistribution structure **160** faces the active surface (including die connectors **126**) of the die **114** and electrically connected to the die **114** through the die connectors **126**. Accordingly, the die **114** can be electrically connected to the functional pads **106** of the backside redistribution structure **110** through, for example, the through vias **112** and the front side redistribution structure **160**. As illustrated in FIG. **7**, the front side redistribution structure **160** includes dielectric layers **132**, **140**, **148**, and **156** and metallization patterns **138**, **146**, and **154**. The front side redistribution structure **160** may be substantially similar to the backside redistribution structure **110** both in formation process and composition. For example, the front side redistribution structure **160** may include one or more layers of dielectric materials (e.g., dielectric layers **132**, **140**, **148**, and **156**) with conductive features (e.g., metallization patterns **138**, **146**, and **154**) formed therein. The one or more dielectric layers may be formed of any suitable material (e.g., polyimide (PI), polybenzoxazole (PBO), BCB, epoxy, silicone, acrylates, nano-filled pheno resin, siloxane, a fluorinated polymer, polynorbornene, an oxide (e.g., silicon oxide), a nitride (e.g., silicon nitride), PSG, BSG, BPSG, or the like) using any suitable method (e.g., a spin-on coating technique, lamination, CVD, sputtering, or the like). The formation of the front side redistribution structure **160** may include patterning the dielectric layers using, for example, photolithography and/or etching processes, and forming conductive features in the patterned dielectric layers by, for example, forming a conductive material in the openings of the patterned dielectric layers.

(30) As an example to form metallization pattern **138**, a seed layer (not shown) is formed over the dielectric layer **132** and in openings through the dielectric layer **132**. In some embodiments, the seed layer is a metal layer, which may be a single layer or a composite layer comprising a plurality of sub-layers formed of different materials. In some embodiments, the seed layer comprises a titanium layer and a copper layer over the titanium layer. The seed layer may be formed using, for example, PVD or the like. A photo resist is then formed and patterned on the seed layer. The photo resist may be formed by spin coating or the like and may be exposed to light for patterning. The

pattern of the photo resist corresponds to the metallization pattern **138**. The patterning forms openings through the photo resist to expose the seed layer. A conductive material is formed in the openings of the photo resist and on the exposed portions of the seed layer. The conductive material may be formed by plating, such as electroplating or electroless plating, or the like. The conductive material may comprise a metal, like copper, titanium, tungsten, aluminum, or the like. Then, the photo resist and portions of the seed layer on which the conductive material is not formed are removed. The photo resist may be removed by an acceptable ashing or stripping process, such as using an oxygen plasma or the like. Once the photo resist is removed, exposed portions of the seed layer are removed, such as by using an acceptable etching process, such as by wet or dry etching. The remaining portions of the seed layer and conductive material form the metallization pattern **138** and vias. The vias are formed in openings through the dielectric layer **132** to, e.g., the through vias **112** and/or the die connectors **126**.

(31) It is noted that the front side redistribution structure **160** is shown as an example. More or fewer dielectric layers and metallization patterns may be formed in the front-side redistribution structure **160**. If fewer dielectric layers and metallization patterns are to be formed, steps and process discussed above may be omitted. If more dielectric layers and metallization patterns are to be formed, steps and processes discussed above may be repeated. One having ordinary skill in the art will readily understand which steps and processes would be omitted or repeated.

(32) Referring to FIG. **8**, the dielectric layer **156** is then patterned. The patterning forms openings to expose portions of the metallization pattern **154**. The patterning may be by an acceptable process, such as by exposing the dielectric layer **156** to light when the dielectric layer is a photo-sensitive material or by etching using, for example, an anisotropic etch. If the dielectric layer **156** is a photo-sensitive material, the dielectric layer **156** can be developed after the exposure.

(33) Then, pads **162** are formed on an exterior side of the front side redistribution structure **160**. The pads **162** are used to couple to conductive connectors **166** (see FIG. **21**) and may be referred to as under bump metallurgies (UBMs) **162**. In the illustrated embodiment, the pads **162** are formed through openings through the dielectric layer **156** to the metallization pattern **154**. As an example to form the pads **162**, a seed layer (not shown) is formed over the dielectric layer **156**. In some embodiments, the seed layer is a metal layer, which may be a single layer or a composite layer comprising a plurality of sub-layers formed of different materials. In some embodiments, the seed layer comprises a titanium layer and a copper layer over the titanium layer. The seed layer may be formed using, for example, PVD or the like. A photo resist is then formed and patterned on the seed layer. The photo resist may be formed by spin coating or the like and may be exposed to light for patterning. The pattern of the photo resist corresponds to the pads **162**. The patterning forms openings through the photo resist to expose the seed layer. A conductive material is formed in the openings of the photo resist and on the exposed portions of the seed layer. The conductive material may be formed by plating, such as electroplating or electroless plating, or the like. The conductive material may comprise a metal, like copper, titanium, tungsten, aluminum, or the like. Then, the photo resist and portions of the seed layer on which the conductive material is not formed are removed. The photo resist may be removed by an acceptable ashing or stripping process, such as using an oxygen plasma or the like. Once the photo resist is removed, exposed portions of the seed layer are removed, such as by using an acceptable etching process, such as by wet or dry etching. The remaining portions of the seed layer and conductive material form the pads **162**. In the embodiment, where the pads **162** are formed differently, more photo resist and patterning steps may be utilized.

(34) Then, conductive connectors **166** are formed on the UBMs **162**. The conductive connectors **166** may be BGA connectors, solder balls, metal pillars, controlled collapse chip connection (C4) bumps, micro bumps, electroless nickel-electroless palladium-immersion gold technique (ENEPIG) formed bumps, or the like. The conductive connectors **166** may include a conductive material such as solder, copper, aluminum, gold, nickel, silver, palladium, tin, the like, or a combination thereof.

In some embodiments, the conductive connectors **166** are formed by initially forming a layer of solder through such commonly used methods such as evaporation, electroplating, printing, solder transfer, ball placement, or the like. Once a layer of solder has been formed on the structure, a reflow may be performed in order to shape the material into the desired bump shapes. In another embodiment, the conductive connectors **166** are metal pillars (such as a copper pillar) formed by a sputtering, printing, electro plating, electroless plating, CVD, or the like. The metal pillars may be solder free and have substantially vertical sidewalls. In some embodiments, a metal cap layer (not shown) is formed on the top of the metal pillar connectors **166**. The metal cap layer may include nickel, tin, tin-lead, gold, silver, palladium, indium, nickel-palladium-gold, nickel-gold, the like, or a combination thereof and may be formed by a plating process.

(35) With now reference to FIG. **9**, a carrier substrate de-bonding is performed to detach (de-bond) the carrier substrate **100** from the backside redistribution structure, e.g., protective layer **104**. In accordance with some embodiments, the de-bonding includes projecting a light such as a laser light or an UV light on the release layer **102** so that the release layer **102** decomposes under the heat of the light and the carrier substrate **100** can be removed. The structure is then flipped over and placed on a tape **190**.

(36) As further illustrated in FIG. **9**, in this orientation, the protective layer **104** is disposed over the redistribution circuit layer. Accordingly, the openings **1041**, **1042** are formed through the protective layer **104** to reveal the functional pads **106** and a part of the dummy pad portions **107** of the redistribution circuit layer. The openings **1041**, **1042** may be formed by using a removal or exposure process such as a laser drilling process. In an embodiment, the laser drilling process may be performed using, e.g., a carbon dioxide (CO₂) laser, although any suitable laser may alternatively be used. During the laser drilling process the drill energy may be in a range from 0.1 mJ to about 30 mJ, and a drill angle of about 0 degree (perpendicular to the protective layer **104**) to about 85 degrees to normal of the protective layer **104**. In some embodiments, the drill time is in a range from about 1 μs to about 150 μs for each of the desired openings **1041**, **1042**. Due to the nature of dummy pads being isolated from other conductive features, the heat applied to the dummy pads is hard to dissipated when undergoing a thermal process such as laser drilling process described above, which would result in damaging the dummy pads and reduce the reliability of the semiconductor package. Accordingly, as illustrated in FIG. **12**, a plurality of dummy pads are integrated into one dummy pad pattern **107** to facilitate the heat dissipation efficiency and reduce the damage to the redistribution circuit layer **110**.

(37) In detail, referring to FIG. **12** to FIG. **14** the dummy pad pattern **107** includes a plurality of pad portions **1071** connected to one another by a connecting portion **1072** in some embodiments. The protective layer **104** includes a plurality of first openings **1041** and a plurality of second openings **1042**. The first openings **1041** are spaced apart from one another and revealing the pad portions **1071** respectively, while the second openings **1042** are spaced apart from one another and revealing the functional pads **106** respectively. In some embodiments, the protective layer **104** covers the connecting portion **1072** and reveals the pad portions **1071** by the first openings **1041**. In one embodiment, the first openings **1041** and the second openings **1042** may be arranged in an array manner, and may be about the same shape and/or size. That is, the shapes of the first openings **1041** remain the same (e.g., circular shape) regardless various different shapes of the dummy pad patterns **107** underneath.

(38) With such arrangement, adjacent (circular) pad portions **1701** are connected to one another by the connecting portion **1702** to form the dummy pad pattern **107**, so that an overall area (footprint) of the dummy pad pattern **107** can be increased, so as to improve the heat dissipation efficiency and reduce damage to the dummy pad pattern **107** from a patterning of the overlying protective layer **104**, such as by laser drilling. In the embodiment shown in FIG. **13** and FIG. **14**, two adjacent pad portions **1701** are connected to each other by the connecting portion **1702** to form the dummy pad pattern **107a**, and the dummy pad pattern **107a** is enclosed by an insulating groove **g1** to isolate the

dummy pad pattern **107a** from other conductive features, e.g., redistribution circuits, functional pads **106**, etc. The (circular) shape of each of the first openings **1041** may be corresponding to the (circular) shape of each of the pad portions **1701**, so the first openings **1041** still define the bonding regions for dummy connectors (e.g., dummy connectors **316**) to be disposed therein in the subsequent process. It is noted that the first openings **1041** are illustrated in dashed line in FIG. **13**, and the first openings **1041** may be the same as or substantially smaller than the corresponding pad portions **1701**.

(39) With now reference to FIG. **15** and FIG. **16**, in the present embodiment, the encapsulated semiconductor device **101** may include at least one dummy through via **113** extending through the encapsulating material **130** and electrically disconnected from the die **114** and the functional through vias **112**. In such arrangement, the location of the dummy through via **113** may at least partially overlap with the dummy pad pattern **107b** from a top view as illustrated in FIG. **15**. Accordingly, the heat applied to the dummy pad pattern **107b** from the thermal process such as laser drilling process may be transmitted to the underlying dummy through via **113**, so as to further facilitate the heat dissipation efficiency and reduce the damage to the dummy pad pattern **107b**. In one embodiment, the dummy through via **113** may be physically disconnected from the dummy pad pattern **107b**, but disposed underneath the dummy pad pattern **107b** for thermal coupling. In alternative embodiment, the dummy through via **113** may be physically connected to the dummy pad pattern **107b** through other dummy features such as dummy vias, dummy lines, etc.

(40) In the present embodiment, the dummy pad pattern **107b** with the dummy through via **113** disposed underneath may be in a circular shape as a normal dummy pad. However, other possible patterns of the dummy pad pattern (e.g., the dummy pad pattern **107a**, **107c**) may also be applied with the dummy through via **113** disposed underneath for further facilitating the heat dissipation efficiency and reducing the damage to the dummy pad pattern. FIG. **15** also illustrates the grounding pad **109** and the functional pad **106** in different forms. In the embodiment, the functional pad **106** may include a pad portion **1061** and a via portion **1062** connecting to the functional vias underneath, and the functional pad **106** is electrically connected to the functional through vias **112**. It is noted that the protective layer **104** is omitted in FIG. **15** for better illustrating the structure underneath.

(41) With now reference to FIG. **17**, in the present embodiment, four adjacent pad portions **1701** are connected to one another by the connecting portion **1702** to form the dummy pad pattern **107c**, and the dummy pad pattern **107c** is enclosed (defined) by an insulating groove **g1** to isolate the dummy pad pattern **107c** from other conductive features. The (circular) shape of each of the first openings **1041** may be corresponding to the (circular) shape of each of the pad portions **1701**, so the first openings **1041** still define the bonding regions for dummy connectors (e.g., dummy connectors **316**) to be disposed therein in the subsequent process. The dummy pad patterns **107c** illustrated above are shown as an example. More or fewer dummy pad portions **1071** may be connected through the connecting portion **1072** to form different dummy pad pattern with different shapes by simply changing the pattern of the photoresist for forming the redistribution circuit layer. It is noted that the protective layer is illustrated in a perspective manner in FIG. **17**, so the first openings **1041** are illustrated in solid line while other structures underneath are illustrated in dashed lines for illustration purpose. In some embodiments, the first openings **1041** may be the same as or substantially smaller than the corresponding pad portions **1701**.

(42) Because the dummy pad pattern **107** and the functional pad **106** within a redistribution circuit layer may be formed simultaneously, the material of the dummy pad pattern **107** and the functional pad **106** may also be the same. In various embodiments, the dummy pad pattern **107** may include a plurality of mesh openings **105** arranged on the connecting portion **1072** and located between the pad portions **1071**. In some embodiments, the mesh openings **105** extend through the dummy pad pattern **107**. For example, the dummy pad pattern **107** may include mesh grids having holes disposed in a grid of rows and columns as illustrated in FIG. **17**. The mesh openings **105** in the

dummy pad pattern **107** may be included to reduce stress induced by the metallization patterns in various dielectric layers of the resulting package. For example, in some embodiments, a total surface area of the mesh openings **105** in a dummy pad pattern **107** may be at least about 30% to about 40% of a total surface of the dummy pattern. It has been observed by including mesh openings **105** having the above areas, stress may be reduced to a suitable level. Thus, manufacturing defects may be reduced and package reliability may be improved. In other embodiments, the mesh openings **105** may occupy a different surface area percentage in relation to a surface area the entire dummy pad pattern **107**.

(43) Referring back to FIG. 9 and FIG. 10, a singulation process is then performed by sawing along scribe line (illustrated as dashed line in FIG. 9), e.g., between adjacent regions **600** and **602**. The sawing process singulates the first package region **600** from the second package region **602**. FIG. 10 illustrates a resulting, singulated package **200**, which may be from one of the first package region **600** or the second package region **602**. The semiconductor package **200** may also be referred to as an integrated fan-out (InFO) package **200**.

(44) Then, referring to FIG. 10 and FIG. 11, a substrate **300** may be mounted on the first package **200** through a plurality of connectors **314**, **316** to form a semiconductor package **500** shown in FIG. 11. The semiconductor package **500** may include the package **200** (may be referred to as a first package **200**), a substrate **300**, and a substrate **400**. In some embodiments, the protective layer **104** with openings (e.g., first openings **1041** and second openings **1042** in FIG. 10) exposes the functional pads **106** and the pad portions **1071** of the dummy pad portions **107**. Then, the connector **314**, **316** may be disposed in the openings **1041**, **1042** and bonded with the functional pads **106** and the dummy pad portions **107** exposed by the openings **1041**, **1042** in accordance with some embodiments. In some embodiments, the connectors includes a plurality of functional connectors **314** and a plurality of dummy connectors **316** electrically disconnected from the functional connectors **314**. The dummy connectors **316** are disposed in the first openings **1041** and connected to the pad portions **1071** of the dummy pad pattern **107** respectively. The functional connectors **314** are disposed in the second openings **1042** and connected to the functional pads **104** respectively.

(45) In some embodiments, the substrate **300** may be fabricated with a predetermined functional circuit. In some embodiments, the substrate **300** may include a variety of electrical circuits suitable for a particular application. In some embodiments, the electrical circuits include various devices such as transistors, capacitors, resistors, diodes and/or the like. In some embodiments, the substrate **300** may be a chip, a die, or a package. In some embodiments, the substrate **300** has a top cross section (a cross section from the top view) in a quadrilateral, a rectangular or a square shape. In some embodiments, the substrate **300** may include an interposer, a redistribution structures, or other mounting surface, with one or more dies disposed thereon. The substrate **300** is in electrical communication with the functional pads **106** through the functional connectors **314**. The connectors **314**, **316** may include, for example, solder balls, conductive bumps, pillars, studs, or another conductive structure. In some embodiments, the substrate **300** may include stacked memory dies. For example, the stacked memory dies may include DRAM, SRAMS, flash memories, low-power (LP) double data rate (DDR) memory modules, such as LPDDR1, LPDDR2, LPDDR3, LPDDR4, or the like memory modules. In other embodiments, the substrate **300** may include any one of various known types of semiconductor devices such as microprocessors, application-specific integrated circuits (ASICs), central computing unit (CPU) or the like.

(46) In some embodiments, before bonding the connectors **314**, **316**, the connectors **314**, **316** are coated with a flux (not shown), such as a no-clean flux. The connectors **314**, **316** may be dipped in the flux or the flux may be jetted onto the connectors **314**, **316**. In another embodiment, the flux may be applied to the surfaces of the functional pads and the pad portions of the dummy pad pattern **107**. The connectors **314**, **316** may have an epoxy flux (not shown) formed thereon before they are reflowed with at least some of the epoxy portion of the epoxy flux remaining after the substrate **300** is attached to the package **200**. This remaining epoxy portion may act as an underfill

to reduce stress and protect the joints resulting from the reflowing the connectors **314**, **316**. In some embodiments, an underfill (not shown) may be formed between the substrate **300** and the package **200** and surrounding the connectors **314**, **316**. The underfill may be formed by a capillary flow process after the substrate **300** is attached or may be formed by a suitable deposition method before the substrate **300** is attached.

(47) In some embodiments, the semiconductor package **500** includes the packages **200** and substrate **300** being mounted to a substrate **400**. The substrate **400** may be referred to a package substrate **400**. The package **200** is mounted to the package substrate **400** using the conductive connectors **166**. The package substrate **400** may be made of a semiconductor material such as silicon, germanium, diamond, or the like. Alternatively, compound materials such as silicon germanium, silicon carbide, gallium arsenic, indium arsenide, indium phosphide, silicon germanium carbide, gallium arsenic phosphide, gallium indium phosphide, combinations of these, and the like, may also be used. Additionally, the package substrate **400** may be a SOI substrate. Generally, an SOI substrate includes a layer of a semiconductor material such as epitaxial silicon, germanium, silicon germanium, SOI, SGOI, or combinations thereof. The package substrate **400** is, in one alternative embodiment, based on an insulating core such as a fiberglass reinforced resin core. One example core material is fiberglass resin such as FR4. Alternatives for the core material include bismaleimide-triazine BT resin, or alternatively, other PCB materials or films. Build up films such as ABF or other laminates may be used for package substrate **400**.

(48) The package substrate **400** may include active and passive devices (not shown in FIG. **11**). As one of ordinary skill in the art will recognize, a wide variety of devices such as transistors, capacitors, resistors, combinations of these, and the like may be used to generate the structural and functional requirements of the design for the semiconductor package **500**. The devices may be formed using any suitable methods.

(49) The package substrate **400** may also include metallization layers and vias (not shown) and bond pads **402** over the metallization layers and vias. The metallization layers may be formed over the active and passive devices and are designed to connect the various devices to form functional circuitry. The metallization layers may be formed of alternating layers of dielectric (e.g., low-k dielectric material) and conductive material (e.g., copper) with vias interconnecting the layers of conductive material and may be formed through any suitable process (such as deposition, damascene, dual damascene, or the like). In some embodiments, the package substrate **400** is substantially free of active and passive devices.

(50) In some embodiments, the conductive connectors **166** can be reflowed to attach the package **200** to the bond pads **402**. The conductive connectors **166** electrically and/or physically couple the substrate **400**, including metallization layers in the substrate **400**, to the package **200**. The conductive connectors **166** may have an epoxy flux (not shown) formed thereon before they are reflowed with at least some of the epoxy portion of the epoxy flux remaining after the package **200** is attached to the substrate **400**. This remaining epoxy portion may act as an underfill to reduce stress and protect the joints resulting from the reflowing the conductive connectors **166**. In some embodiments, an underfill (not shown) may be formed between the first package **200** and the substrate **400** and surrounding the conductive connectors **166**. The underfill may be formed by a capillary flow process after the package **200** is attached or may be formed by a suitable deposition method before the package **200** is attached.

(51) Embodiments of the semiconductor package and methods in the current disclosure have many advantages. In particular, adjacent pad portions **1701** are connected to one another by the connecting portion **1702** to form the dummy pad pattern **107**, so that an overall area (footprint) of the dummy pad pattern **107** can be increased, so as to improve the heat dissipation efficiency and reduce damage to the dummy pad pattern **107** from a patterning of the overlying protective layer **104**, such as by laser drilling. For example, for the embodiment shown in FIG. **15**, highest temperature that the dummy pad pattern **107b** had reached during the laser drilling process is

decreased about 10% to about 15%. For the embodiment shown in FIG. 13, highest temperature that the dummy pad pattern **107a** had reached during the laser drilling process is decreased about 30% to about 35%. For the embodiment shown in FIG. 17, highest temperature that the dummy pad pattern **107c** had reached during the laser drilling process is decreased about 50% to about 55%. Therefore, by providing the dummy structure adjacent to the redistribution layer, damage to the redistribution layer may be reduced from a patterning of an overlying passivation layer, such as by laser drilling. By connecting adjacent pad portions **1701** by the connecting portion **1702** to form the dummy pad pattern **107**, reducing or eliminating the damage to the dummy pad pattern **107** caused by the thermal process may be achieved.

(52) Based on the above discussions, it can be seen that the present disclosure offers various advantages. It is understood, however, that not all advantages are necessarily discussed herein, and other embodiments may offer different advantages, and that no particular advantage is required for all embodiments.

(53) Other features and processes may also be included. For example, testing structures may be included to aid in the verification testing of the 3D packaging or 3DIC devices. The testing structures may include, for example, test pads formed in a redistribution layer or on a substrate that allows the testing of the 3D packaging or 3DIC, the use of probes and/or probe cards, and the like. The verification testing may be performed on intermediate structures as well as the final structure. Additionally, the structures and methods disclosed herein may be used in conjunction with testing methodologies that incorporate intermediate verification of known good dies to increase the yield and decrease costs.

(54) In accordance with some embodiments of the disclosure, a semiconductor package includes a substrate, a redistribution circuit layer, and a protective layer. The redistribution circuit layer is over the substrate and includes a plurality of functional pads electrically connected to the substrate, and a dummy pad pattern electrically disconnected from the plurality of functional pads, wherein the dummy pad pattern includes a plurality of pad portions connected to one another. The protective layer is disposed over the redistribution circuit layer and comprising a plurality of first openings spaced apart from one another and respectively revealing the plurality of pad portions. In an embodiment, the semiconductor package further includes a plurality of dummy connectors disposed in the plurality of first openings and connected to the plurality of pad portions of the dummy pad pattern. In an embodiment, the protective layer further includes a plurality of second openings separated from one another and respectively revealing the plurality of functional pads. In an embodiment, the plurality of first openings and the plurality of second openings are arranged in an array manner. In an embodiment, the semiconductor package further includes a plurality of functional connectors disposed in the plurality of second openings and connected to the plurality of functional pads respectively. In an embodiment, the dummy pad pattern further includes a plurality of mesh openings disposed between the plurality of pad portions. In an embodiment, the substrate includes a die and encapsulating material at least laterally encapsulating the die and a plurality of through vias extending through the encapsulating material and electrically connected to the die. In an embodiment, the semiconductor package further includes a backside redistribution structure disposed over a backside of the die and the encapsulating material, wherein the backside redistribution structure includes the redistribution circuit layer. In an embodiment, the semiconductor package further includes an adhesive disposed on the backside of the die and located between the die and the redistribution structure. In an embodiment, the semiconductor package further includes a front side redistribution structure disposed over a front side of the die and the encapsulating material.

(55) In accordance with some embodiments of the disclosure, a semiconductor package includes an encapsulated semiconductor device, a backside redistribution structure, and a protective layer. The encapsulated semiconductor device includes an encapsulating material encapsulating a die. The backside redistribution structure is disposed over a backside of the encapsulated semiconductor

device and including a redistribution circuit layer, wherein the redistribution circuit layer includes a plurality of functional pads electrically connected to the die, and a dummy pad pattern is electrically disconnected from the plurality of functional pads, wherein the dummy pad pattern includes a plurality of pad portions and a connecting portion connecting the plurality of pad portions. The protective layer is disposed over the redistribution circuit layer, wherein the protective layer covers the connecting portion and reveals the plurality of pad portions and the plurality of functional pads. In an embodiment, the die includes an active surface having a plurality of die connectors and a back surface opposite to the active surface, and the backside redistribution structure faces the back surface of the die. In an embodiment, the semiconductor package further includes a front side redistribution structure disposed over a front side of the encapsulated semiconductor device and facing the active surface of the die. In an embodiment, the encapsulated semiconductor device includes a plurality of functional through vias extending through the encapsulating material and electrically connected to the die. In an embodiment, the dummy pad pattern further includes a plurality of mesh openings arranged on the connecting portion and between the plurality of pad portions.

(56) In accordance with some embodiments of the disclosure, a semiconductor package includes an encapsulated semiconductor device and a redistribution structure. The encapsulated semiconductor device includes an encapsulating material encapsulating a die, a dummy through via extending through the encapsulating material and electrically disconnected from the die. The redistribution structure is disposed over the encapsulated semiconductor device and includes a redistribution circuit layer, wherein the redistribution circuit layer includes a plurality of functional pads electrically connected to the die, and a dummy pad pattern electrically disconnected from the plurality of functional pads, wherein the dummy through via at least partially overlap with the dummy pad pattern from a top view. In an embodiment, the semiconductor package further includes a plurality of functional through vias extending through the encapsulating material, wherein the plurality of functional through vias are electrically connected to the die and electrically disconnected from the dummy through via. In an embodiment, the semiconductor package further includes a protective layer disposed over the redistribution circuit layer and including a plurality of openings revealing the dummy pad pattern and the plurality of functional pads. In an embodiment, the semiconductor package further includes a plurality of dummy connectors connected to the plurality of the dummy pad pattern and a plurality of functional connectors connected to the plurality of functional pads. In an embodiment, the semiconductor package further includes an adhesive attached to a backside of the die and disposed between the die and the redistribution structure.

(57) The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A semiconductor package, comprising: a substrate; a redistribution structure over the substrate and comprising a plurality of functional pads electrically connected to the substrate and laterally encapsulated by a dielectric layer, and a dummy pad pattern electrically disconnected from the plurality of functional pads and laterally encapsulated by the dielectric layer, wherein the dummy pad pattern comprises a plurality of pad portions and a connecting portion connecting the plurality

of pad portions; and a protective layer disposed over the dielectric layer, wherein the protective layer covers the connecting portion and comprises a plurality of first openings spaced apart from one another and respectively revealing the plurality of pad portions.

2. The semiconductor package as claimed in claim 1, further comprising a plurality of dummy connectors disposed in the plurality of first openings and connected to the plurality of pad portions of the dummy pad pattern.

3. The semiconductor package as claimed in claim 1, wherein the protective layer further comprises a plurality of second openings separated from one another and respectively revealing the plurality of functional pads.

4. The semiconductor package as claimed in claim 3, wherein the plurality of first openings and the plurality of second openings are arranged in an array manner.

5. The semiconductor package as claimed in claim 3, further comprising a plurality of functional connectors disposed in the plurality of second openings and connected to the plurality of functional pads respectively.

6. The semiconductor package as claimed in claim 1, wherein the dummy pad pattern further comprises a plurality of mesh openings disposed between the plurality of pad portions.

7. The semiconductor package as claimed in claim 1, wherein the substrate comprises a die and encapsulating material at least laterally encapsulating the die and a plurality of through vias extending through the encapsulating material and electrically connected to the die.

8. The semiconductor package as claimed in claim 7, further comprising a backside redistribution structure disposed over a backside of the die and the encapsulating material, wherein the backside redistribution structure comprises the redistribution circuit layer.

9. The semiconductor package as claimed in claim 7, further comprising an adhesive disposed on the backside of the die and located between the die and the redistribution structure.

10. The semiconductor package as claimed in claim 7, further comprising a front side redistribution structure disposed over a front side of the die and the encapsulating material.

11. A semiconductor package, comprising: an encapsulated semiconductor device comprising an encapsulating material encapsulating a die; a backside redistribution structure disposed over a backside of the encapsulated semiconductor device and comprising a redistribution circuit layer, wherein the redistribution circuit layer comprises a plurality of functional pads electrically connected to the die, and a dummy pad pattern electrically disconnected from the plurality of functional pads, wherein the dummy pad pattern comprises a plurality of pad portions, a connecting portion connecting the plurality of pad portions, and a plurality of mesh openings arranged on the connecting portion and located between the pad portions; and a protective layer disposed over the dielectric layer, wherein the protective layer covers the connecting portion and reveals the plurality of pad portions and the plurality of functional pads.

12. The semiconductor package as claimed in claim 11, wherein the die comprises an active surface having a plurality of die connectors and a back surface opposite to the active surface, and the backside redistribution structure faces the back surface of the die.

13. The semiconductor package as claimed in claim 12, further comprising a front side redistribution structure disposed over a front side of the encapsulated semiconductor device and facing the active surface of the die.

14. The semiconductor package as claimed in claim 11, wherein the encapsulated semiconductor device comprises a plurality of functional through vias extending through the encapsulating material and electrically connected to the die.

15. A semiconductor package, comprising: an encapsulated semiconductor device comprising an encapsulating material encapsulating a die, a dummy through via extending through the encapsulating material and electrically disconnected from the die; and a redistribution structure disposed over a backside of the encapsulated semiconductor device and comprising a dielectric layer and a redistribution circuit layer, wherein the redistribution circuit layer comprises a plurality

of functional pads electrically connected to the die, and a dummy pad pattern electrically disconnected from the plurality of functional pads, wherein the dummy through via at least partially overlap with the dummy pad pattern from a top view, and the dummy pad pattern is isolated from the dummy through via by the dielectric layer.

16. The semiconductor package as claimed in claim 15, further comprising a plurality of functional through vias extending through the encapsulating material, wherein the plurality of functional through vias are electrically connected to the die and electrically disconnected from the dummy through via.

17. The semiconductor package as claimed in claim 15, further comprising a protective layer disposed over the redistribution circuit layer and comprising a plurality of openings revealing the dummy pad pattern and the plurality of functional pads.

18. The semiconductor package as claimed in claim 15, further comprising a plurality of dummy connectors connected to the plurality of the dummy pad pattern and a plurality of functional connectors connected to the plurality of functional pads.

19. The semiconductor package as claimed in claim 15, further comprising an adhesive attached to a backside of the die and disposed between the die and the redistribution structure.

20. The semiconductor package as claimed in claim 11, wherein a total surface area of the plurality of mesh openings is substantially equal to or greater than 30% of a total surface area of the dummy pad pattern.
